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College of Osteopathic Medicine

Clinical: Urinary Tract Infections

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Principles and Practice of Osteopathic Medicine 2

By the end of this session, students will be able to…

- Identify the clinical findings/presentations when a patient has a urinary tract infection
- Develop a differential diagnosis in regard to infections of the urinary tract
- Develop a plan for ordering tests using evidence-based guidance and understand how to interpret those laboratory tests
- Develop a treatment plan for patients with a specific urinary tract infections
- Identify patient populations that are at specific risk of adverse events when an infection of the urinary system occurs

Source: Course Syllabus

Definitions

Urinary Tract Infection (UTI)

- Any infection involving the urothelium
  - Asymptomatic bacteriuria (ASB)
  - Urethritis
  - Cystitis
  - Prostatitis
  - Pyelonephritis

Uncomplicated UTI

- Infection in a structurally and neurologically normal urinary tract.

Complicated UTI

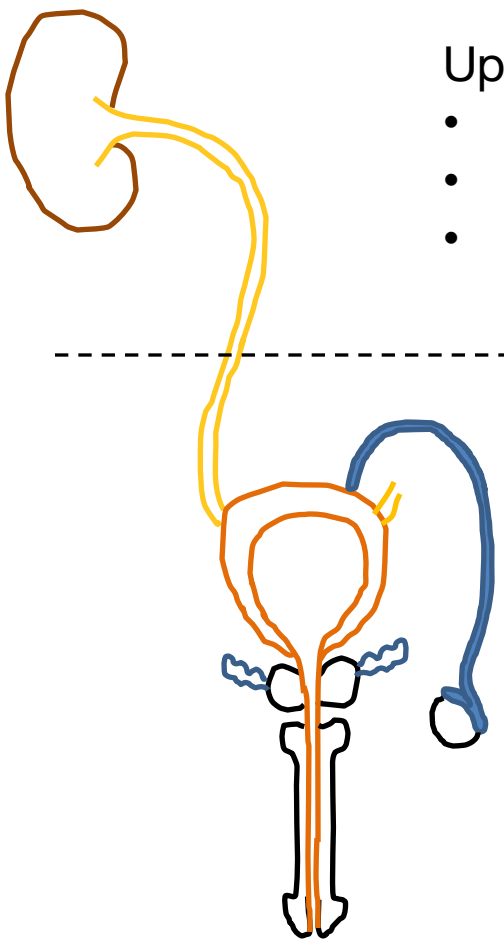
- Infection associated with an underlying condition that increases the risk of failing therapy

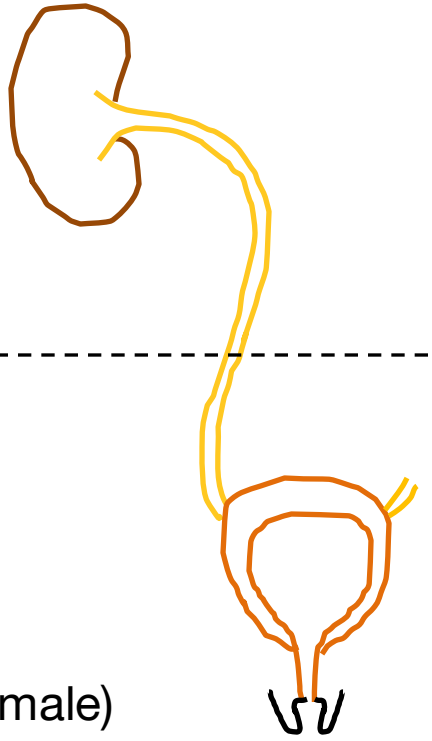
Urinary Tract Infections, Cystitis, Prostatitis, and Pyelonephritis, *Harrison's Principles of Internal Medicine, 22e*

Anatomic Classification

Male

Female





Upper UTI:

- Pyelonephritis
- Intra-renal abscess
- Perinephric abscess

Lower UTI:

- Cystitis
- Urethritis
- Prostatitis (male)

Risk Factors by Age Group and Pathogenesis

Age in Years	Females	Males
< 1	Anatomic or functional urologic abnormalities	
1-5	Congenital abnormalities Vesicoureteral reflux	Congenital abnormalities Uncircumcised penis*
6-15	Vesicoureteral reflux	
16-35	Sexual intercourse Diaphragm use Spermicide use Previous UTI	Anatomic urologic abnormality Rectal intercourse
36-65	Gynecologic surgery Bladder prolapse Previous UTI	Prostate hypertrophy Obstruction Catheterization Surgery
>65	Estrogen deficiency and loss of vaginal flora	All of the above Incontinence Long-term catheterization

Comparison of Symptoms by Age Group

- Newborns
  - Fever or Hypothermia
  - Decreased feeding
- Infants
  - Vomiting / Diarrhea
  - Fever
  - Decreased feeding or failure to thrive
- Children
  - Irritability
  - Change in urinary habits
  - Poor appetite
- Elderly
  - Fever or Hypothermia
  - Poor appetite
  - “Change in Mental Status”

Epidemiology

- Increased risk in:
  - Neonates (male > female)
  - Girls/ young women
  - Older men
  - Individuals with structural abnormalities of the urinary tract
  - Immunosuppressed

Asymptomatic Bacteriuria

- Absence of symptoms attributable to the bacteria in the urinary tract and does not usually require treatment.
- Incidentally found on screening urine culture.

Diagnosis:

- Women:
  - Clean-catch voided specimens on two separate occasions must contain $>10^5$ CFU/mL of the same bacterial strain, **or**
  - One catheterized specimen must contain $>10^2$ CFU/mL of bacteria.
- Men:
  - A single clean-catch specimen with $>10^5$CFU/mL of bacteria **or**
  - One catheterized specimen with $>10^2$ CFU/mL of bacteria.

Urinary Tract Infections, Cystitis, Prostatitis, and Pyelonephritis, *Harrison's Principles of Internal Medicine, 22e*

Asymptomatic Bacteriuria

- **Screen and treat pregnant women** (12-16 weeks gestation)
  - 7-day course of amoxicillin
  - Penicillin allergy: 7-day course of Nitrofurantoin or a cephalosporin

Case #1

A 29-year-old woman complains of a 2-day history of dysuria, urgency, and urinary frequency. She denies the use of medications and has no significant past medical history. On examination, her blood pressure (BP) is 100/70 mm Hg, heart rate (HR) is 90 beats per minute, and temperature is 98°F (36.6°C). The heart and lung examinations are normal. She does not have back tenderness. The abdomen is nontender and without masses. The pelvic examination reveals normal female genitalia. There is no adnexal tenderness or masses.

What is the most likely diagnosis?

What is the next step in the diagnosis?

Uncomplicated Bacterial Cystitis

- Most common in women

- Diagnosis
  - Dysuria
  - Frequency, urgency, or both
  - Cloudy, smelly urine
  - Urine dipstick analysis positive for nitrites or leukocyte esterase.
  - Positive urine culture (>10^4 organisms).
  - **No fever or flank pain**

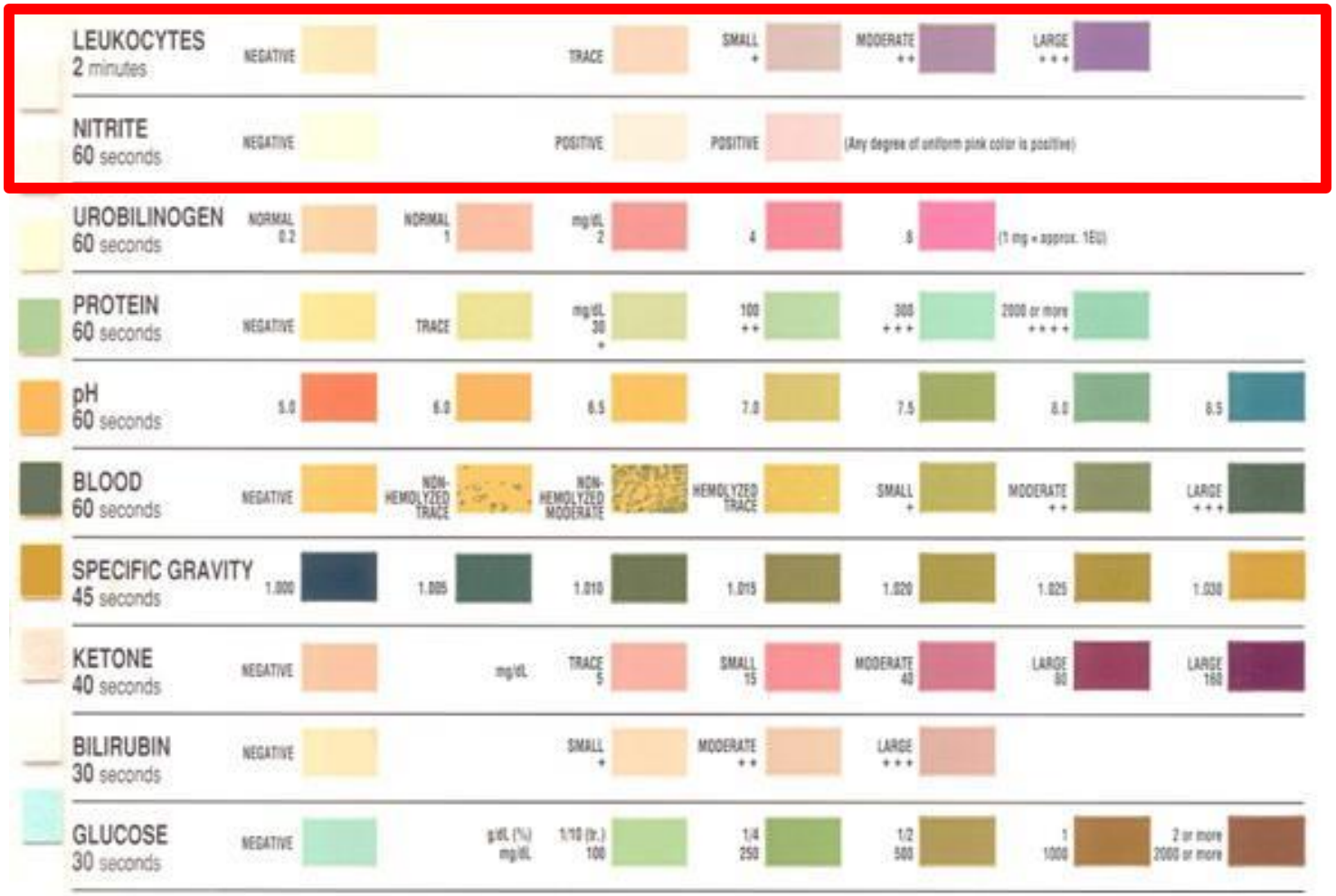
Uncomplicated Bacterial Cystitis [Young Women]

Factors that promote cystitis	Factors that prevent cystitis	Factors with no evidence of effect on cystitis
• Sexual activity and practices • Genetic predisposition • Delayed postcoital voiding	• Cranberry juice • Prophylactic antibiotics	• Precoital voiding • Underwear fabric • Douching • Hot tub use

Diagnostic Testing

- Urinalysis (in office dip-stick or laboratory analysis)
- Urine Culture
- Complete Blood Cell Count
- Basic Metabolic Profile
- Imaging Studies

Urinalysis



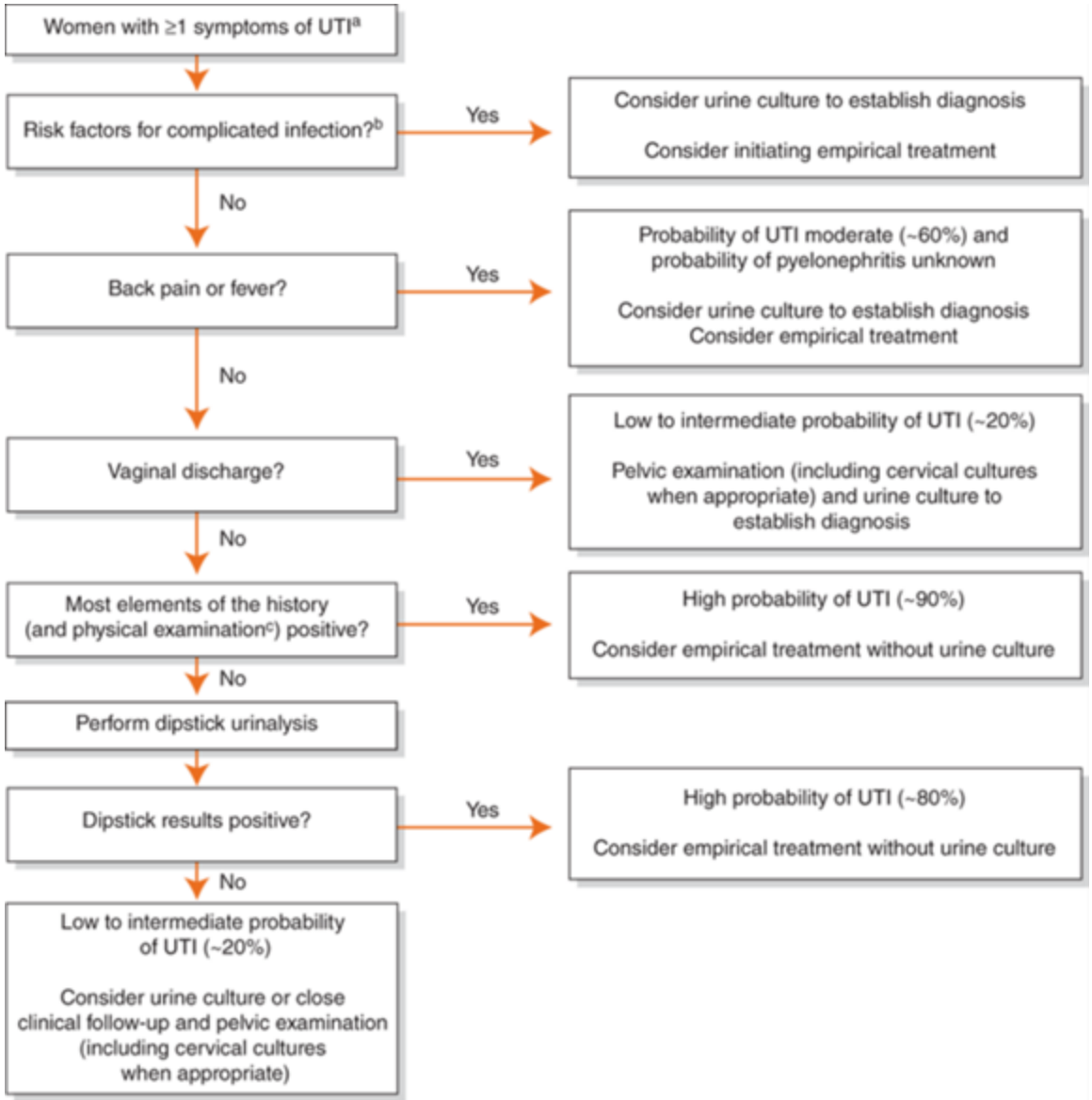
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Basic guidelines – Treatment of UTI

- Treatment is directly related to level of infection
- In most females, treatment may be started without testing
- Any other case, urine culture must be done. Use empirical antibiotics while waiting
- In male and children, follow-up investigation must be done after first UTI
- Differential diagnosis always includes STI

Diagnostic Algorithm for Cystitis



Source: J. E. South-Paul, S. C. Matheny, E. L. Lewis: CURRENT Diagnosis & Treatment: Family Medicine, 4th Ed.
www.accessmedicine.com
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Acute Cystitis - Treatment

Uncomplicated

- **Trimethoprim-sulfamethoxazole** x **3** days

Complicated

- **TMP/SMX** Long course (**≥ 10-14** days)

Sulfa allergy

- **Nitrofurantoin** x **5-7** days or
- **Fluoroquinolone** x **3** days
  - Levofloxacin
  - Ofloxacin
  - Ciprofloxacin

Pregnant women

- **Amoxicillin**, **cephalosporin**, **nitrofurantoin**, or another pregnancy-safe antibiotic x **7** days

UTI in Pregnancy

- 2-8% of pregnant women will have a UTI
- 20-30% with asymptomatic bacteriuria develop pyelonephritis
- Physiologically caused by:
  - Decreased ureteral tone and peristalsis
  - Incompetence of vesicoureteral valve
- UTI in pregnancy leads to increased risk:
  - Low-birth weight baby
  - Premature delivery
  - Newborn mortality

ALL UTIs (symptomatic and asymptomatic) MUST be treated during pregnancy!

Case #2

An 18-year-old male presents to the clinic complaining of a burning sensation during urination.

He also reports pruritis and purulent discharge at the urethral meatus. He recently had unprotected sex with a new partner.

Urethritis

Diagnosis

- Pain or irritation on urination
- No frequency or urgency
- No or low level suprapubic tenderness
- + Discharge from the urethra (predominantly males)
- Vaginal discharge possible
- Low bacterial count or “sterile urine” on culture

E. coli UTI favored if:

- Gross hematuria
- Abrupt onset
- Duration < 3 days
- Hx of previous UTI

Rule out STI

Case #3

A 51-year-old man presented with fever and general malaise of 2 week duration. He has had diabetes mellitus for the preceding 20 years, and at the time of presentation this condition was poorly controlled.

On admission to the hospital, his white-cell count was 10,800 per cubic millimeter (normal range, 3900 to 9800). The C-reactive protein level was 8.6 mg/dl, blood urea nitrogen 90 mg per deciliter (10-30 mg/dl), creatinine 4.9 mg per deciliter (1-2 mg/dl), and glycated hemoglobin 11.2%.

Acute Pyelonephritis

- Clinical Findings:
  - Mild: low-grade fever without or with lower-back or costovertebral angle (CVA) tenderness
  - Severe high fever, rigors, nausea, vomiting, flank and/or loin pain
  - **Fever** is the main feature distinguishing cystitis and pyelonephritis

- Diabetic patients: obstructive uropathy associated with acute papillary necrosis, emphysematous pyelonephritis

Acute Pyelonephritis

- May be managed as outpatient
- Hospitalize if:
  - Intractable vomiting
  - Evidence of shock
  - Severe dehydration
- Initiate treatment in the hospital for pregnant females
- Treatment should last at least 10 days
- Follow-up 5 days after treatment completion

Acute Pyelonephritis - Treatment

- Obtain urine culture, then start empiric treatment

- Bactericidal, broad spectrum
  - Fluoroquinolones
  - Aminoglycosides
  - Aminopenicillins (amoxicillin with clavulanic acid, ticarcillin, or piperacillin)
  - Cephalosporins
  - Imipenem

- Treat for 10 days of coverage
  - May need to start IV antibiotics if PO not tolerated

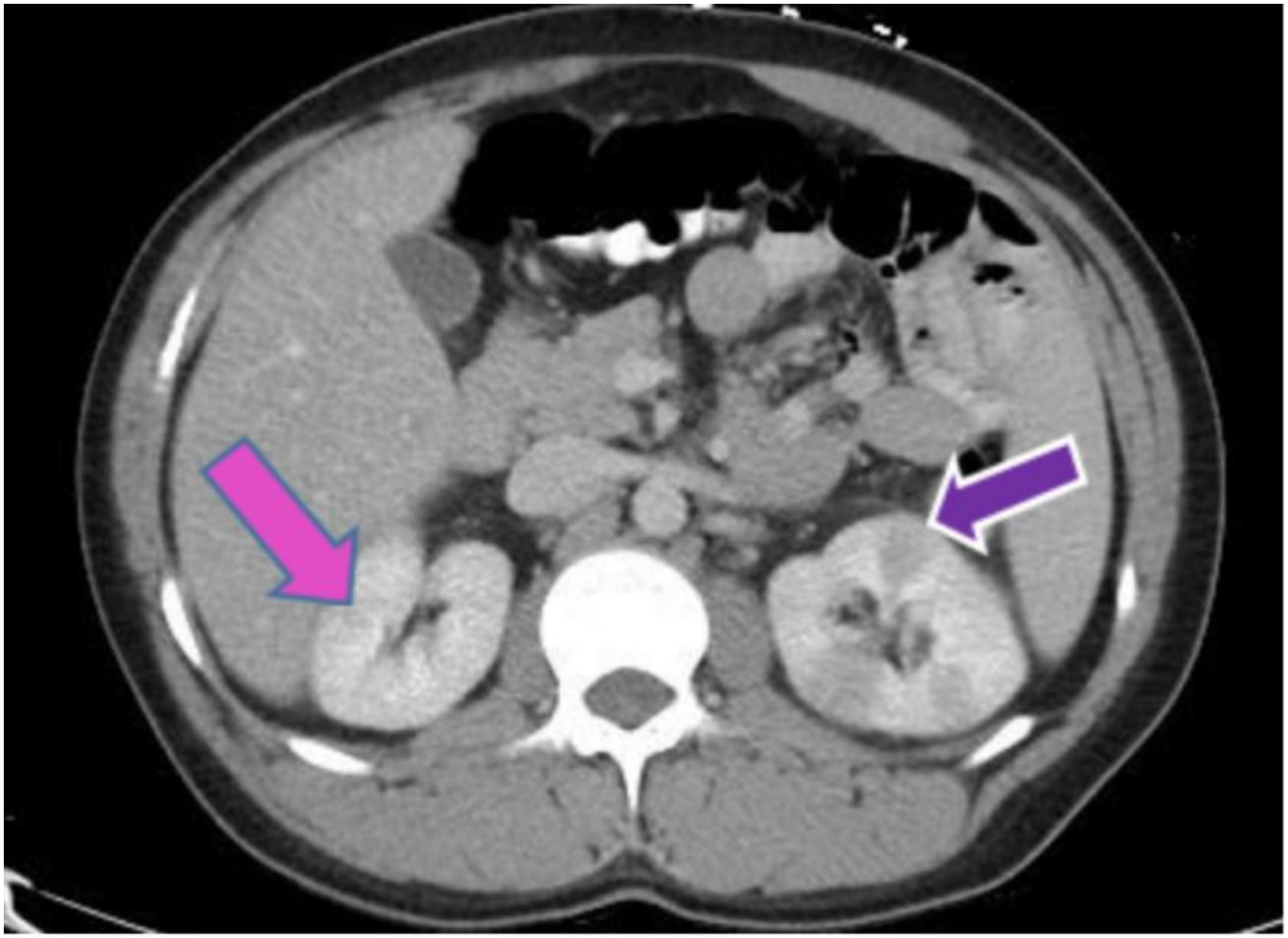
Acute Pyelonephritis - Treatment

- Special Considerations in Pregnancy
  - Initiate treatment with IV or IM in hospital
  - When afebrile, patient may take oral meds to complete 10 days treatment
  - Ensure eradication of organism with repeat urine culture 5 days post treatment

Pyelonephritis

- Indications for imaging
  - Uncertain diagnosis
  - Severe illness or immunocompromised status
  - Worsening renal function
  - Suspected kidney stone or urinary tract obstruction
  - Relapsing or recurrent pyelonephritis
  - Male gender
  - Failure to improve after 72 hours of antibiotics

CT finding of pyelonephritis



Case #4

A 63-year-old male presents to the ED with a 2-day history of fever, urinary frequency, dysuria, and difficulty initiating the urinary stream.

He also relates having some perineal pain.

On examination, his vitals are stable except for a temperature of 101.3°F.

Consensus Classification of Prostatitis

Syndrome	Description	WBC count per HPF (urine)	Bacterial Cultures	Special Tests
Acute Bacterial Prostatitis	Acute infection of the prostate gland	>10	+ on voided specimen	
Chronic Bacterial Prostatitis	Chronic infection of the prostate gland	>10	(-) on voided specimen	Expressed prostatic secretions will be +
Chronic Pelvic Pain Syndrome	Chronic pelvic pain in the absence of bacteria localized to the prostate	>10 (In non-inflammatory causes there will be <10)	(-) bacteria on any specimen	

Acute Bacterial Prostatitis

Symptoms:

- Dysuria, frequency, and urgency
- Low back, perineal, penile, or rectal pain
- Fever and chills

Physical Exam:

- Tense/boggy prostate
- Purulent discharge
- Urine culture positive
  - Most commonly gram-negative organism (*E. coli or Klebsiella)*

Acute Bacterial Prostatitis

Treatment:

- Quinolones (ciprofloxacin or norfloxacin) - Treatment of choice if not contraindicated
- TMP/SMX (Check sensitivities prior due to resistance)
- Duration of treatment: 2-4 weeks

Chronic Prostatitis

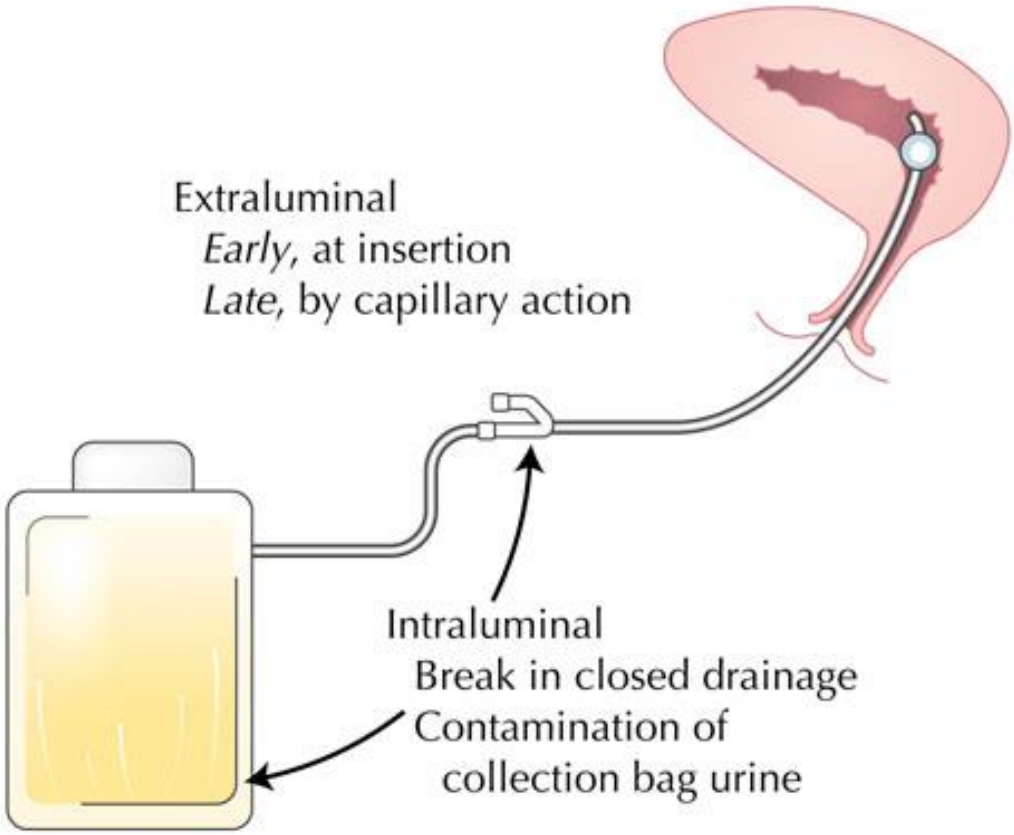
- 70% of cases have “sterile urine”
- Culture urine and treat initially
- If there is pyuria “without bacteria”
  - Test for Chlamydia

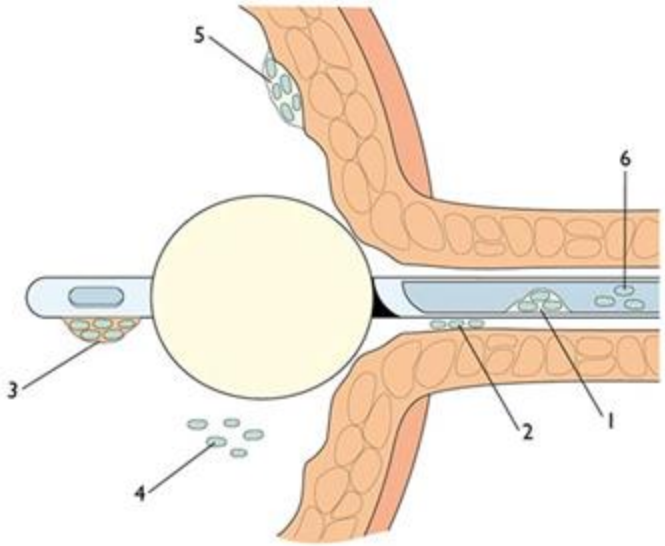
Case #5

A 22-year-old female presents to the ED with 3 days of increased urinary frequency and suprapubic pain after micturition. She has no fevers, chills, or flank pain, nausea or vomiting, and urethral discharge.

There have been no similar complaints in the past. She is sexually active and uses a barrier mode of contraception, however remarks that LMP was 3 months prior.

Iatrogenic – Catheter Induced





Nosocomial Infections in the Intensive Care Unit Associated with Invasive Medical Devices. Current Infectious Disease Reports. 3(6):487-495.

Complications from foley catheter insertion

* good to know during clinical rotations

Common

- Asymptomatic bacteriuria
- Urinary tract infection
- Hematuria from traumatic insertion/bladder irritation
- Bladder spasm
- Diminished patient mobility
- Encrustation and blockage by bacterial biofilm
- Retained catheter due to not deflating balloon

Uncommon

- Urethral false passage
- Balloon inflation in urethra
- Urethral stricture or necrosis
- Bladder stone formation around the balloon
- Polypoid cystitis
- Bladder perforation or peritonitis
- Laceration or erosion of the glans penis

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Reference: Principles and Practice of Hospital Medicine, Chapter 205

Nursing Homes/ Long Term Care Settings

Prevalence of UTI reported 15-50% in LTC settings

- Likely to be higher than current estimates

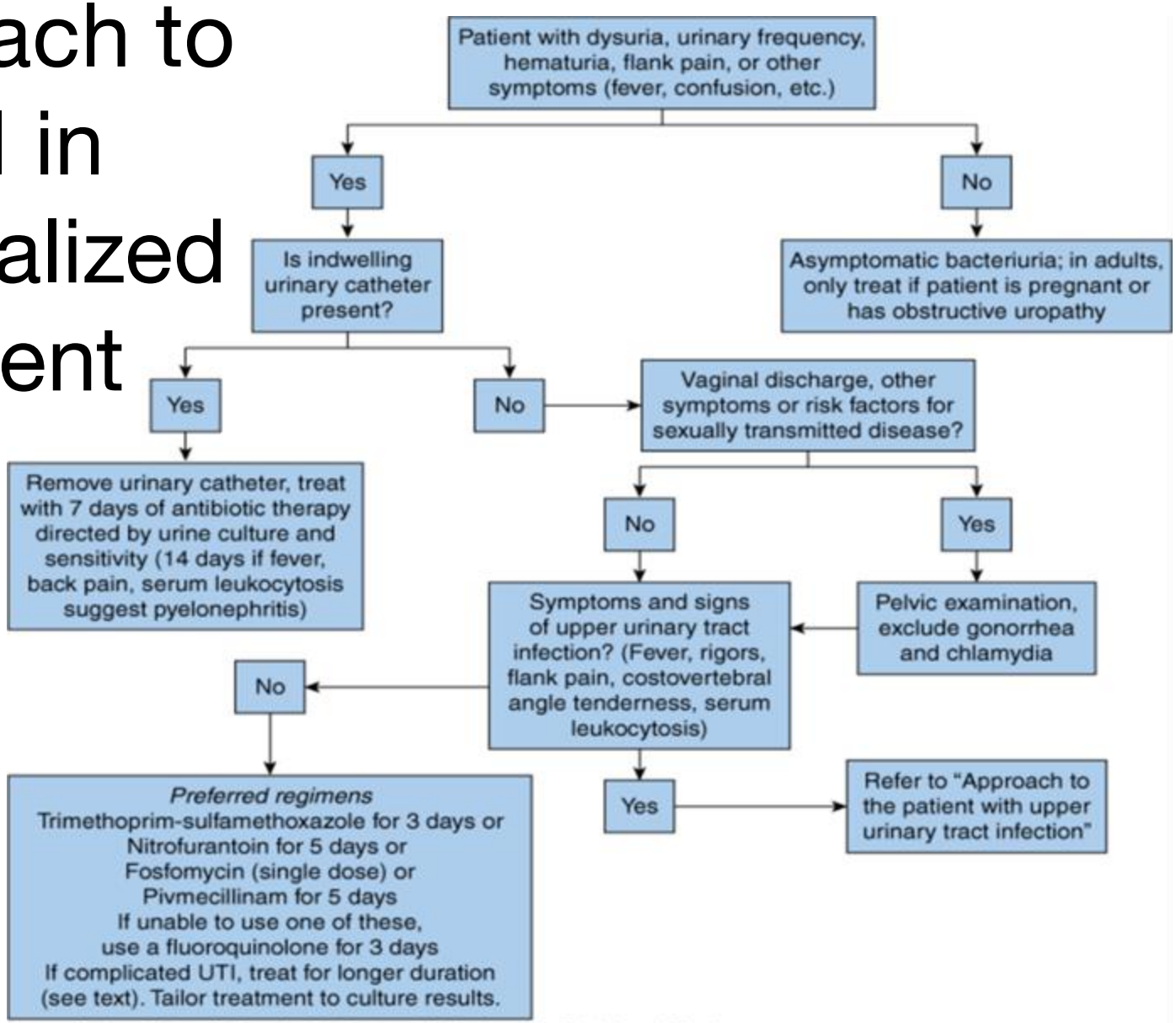
Special Issues

- Multidrug resistant bacteria and overgrowth of organisms such as *C difficile*
- *Pseudomonas, enterococci, staphylococci*, and *candida* species are more common than in the outside population
- E. coli is seen in 50% (however tend to be resistant strains)

Use urine culture and sensitivity

Treat using narrow spectrum antibiotics

Approach to UTI in hospitalized patient

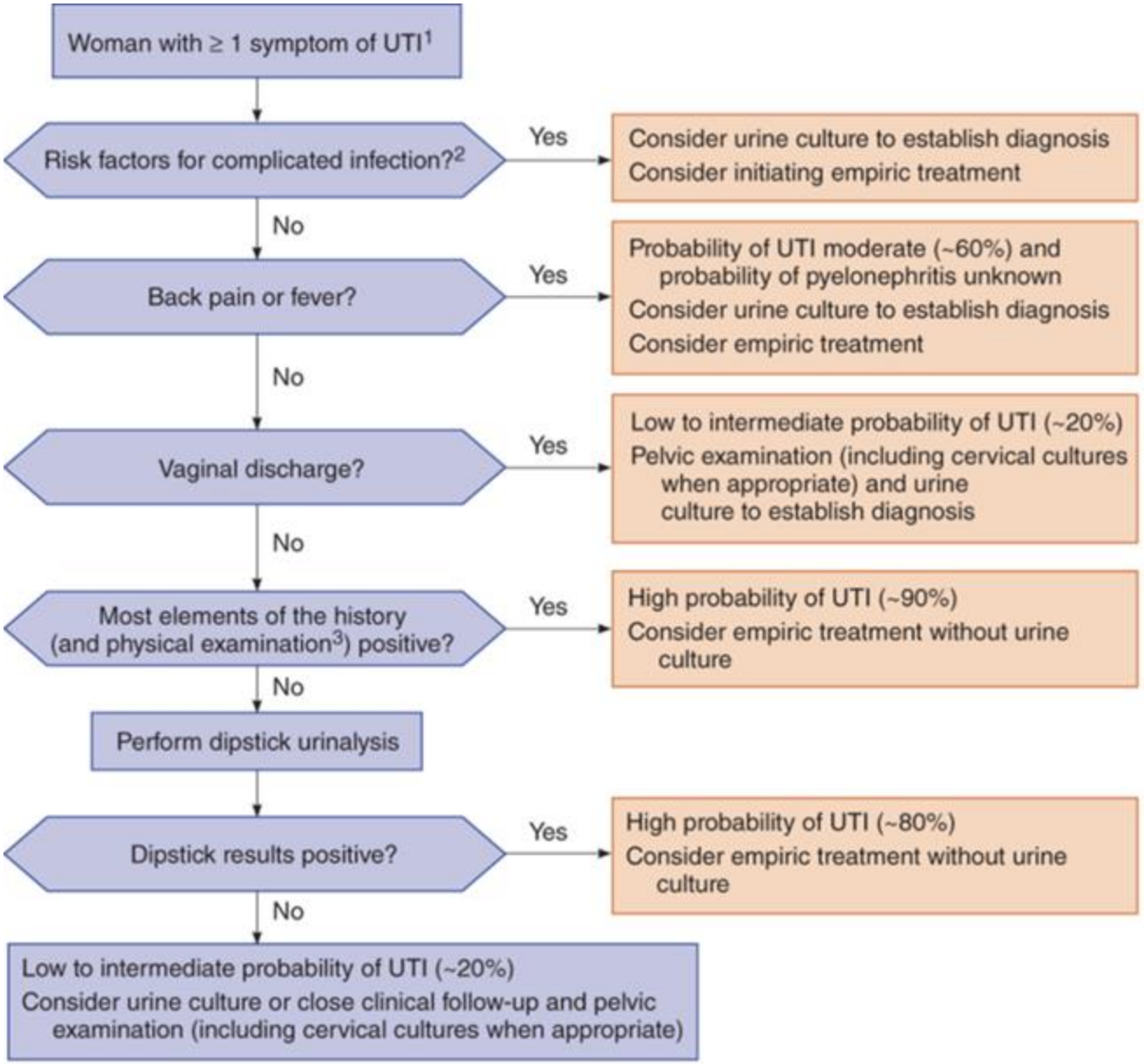


Source: McKean S, Ross JJ, Dressler DD, Brotman DJ, Ginsberg JS: *Principles and Practice of Hospital Medicine*: www.accessmedicine.com

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Approach to UTI for the female patient



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Reference: from Bent S et al. Does this woman have an acute uncomplicated urinary tract infection? JAMA. 2002 May 22–29;287(20):2701–10. © 2002 American Medical Association

Clinical decision making for:

When to refer to a specialist:

- Anatomic abnormalities leading to repeated urinary infections
- Infections associated with nephrolithiasis
- Persistent interstitial cystitis/painful bladder syndrome

When to admit the non-pregnant patient:

- Severe pain requiring parenteral medication or impairing ambulation or urination
- Dysuria associated with urinary retention or obstruction
- Pyelonephritis with ureteral obstruction

Summary - Key Questions to Consider

- Which patients with urinary tract infections need hospital admission?
- When should catheter-associated urinary tract infections be treated?
- When is renal imaging indicated?
- When should urology consultation be obtained?
- What are the appropriate durations of therapy for uncomplicated cystitis, catheter-associated urinary tract infections, and pyelonephritis?
- For which patients are follow-up urine cultures indicated after discharge?

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

